



Software Engineering Department
Braude College of Engineering

Final Project – Phase B (61998)

**Robust Remote Heart Rate Estimation under
Artificial Lighting Conditions**

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1 Abstract

Remote photoplethysmography (rPPG) enables contactless heart rate measurement by analyzing subtle color variations in facial skin captured through a standard camera. While this technique yields promising results under natural daylight, its reliability significantly deteriorates in indoor environments under artificial lighting due to flickering, uneven illumination, and spectral distortions that obscure pulse-related signal components.

To address these limitations, this project investigated and integrated advanced signal and image processing techniques designed to suppress lighting-induced noise. Specifically, we evaluated several algorithmic approaches, including chrominance-based projection methods such as POS and CHROM, together with Retinex filtering for illumination normalization. This report presents a comprehensive analysis of the results, assessing the performance, relative impact, and efficacy of each algorithm in mitigating artificial lighting noise to achieve robust indoor heart rate estimation using standard laptop webcam video streams.

2 Introduction

Accurate heart rate monitoring is a fundamental component of modern healthcare, particularly for elderly individuals in nursing homes or remote-care environments where continuous contact-based monitoring may be impractical or unavailable. In recent years, contactless pulse detection through remote photoplethysmography (rPPG) has attracted substantial attention due to its simplicity, accessibility, and potential for seamless integration into camera-based monitoring systems.

A previous project developed a foundational rPPG framework that isolated the subject’s forehead region from a video stream and applied frequency-domain analysis to the green color channel in order to estimate pulse rate. While this baseline architecture demonstrated promising accuracy under outdoor conditions and stable natural daylight, its reliability deteriorated significantly under indoor artificial illumination. Signal instability, spectral artifacts, and illumination-induced noise generated by artificial light sources—particularly LED and fluorescent lighting—interfered with the extraction of the true photoplethysmographic signal and limited the practical applicability of the system.

To address these limitations, the present project extended the original framework into a more robust and research-driven physiological monitoring system through the investigation and implementation of advanced signal and image-processing strategies. Rather than relying exclusively on single-channel analysis, the project gradually transitioned toward chrominance-based physiological modelling, primarily through the Plane-Orthogonal-to-Skin (POS) algorithm, while also evaluating the Chrominance-based (CHROM) method as a comparative approach. In parallel, illumination normalization strategies inspired by Retinex theory were investigated to address the interaction between lighting conditions and skin reflectance.

The spatial tracking subsystem was simultaneously redesigned using an OpenCV Haar

Cascade face detector backbone paired with localized, deterministic pixel offsets to enable stable forehead Region of Interest (ROI) extraction under varying motion and acquisition conditions. The signal-processing pipeline was further enhanced through normalized RGB extraction, temporal detrending, Butterworth band-pass filtering, Blackman windowing, FFT-based frequency analysis, and runtime signal-quality validation mechanisms.

Throughout development, considerable effort was devoted to balancing illumination normalization, spectral sensitivity, and temporal stability, particularly under webcam acquisition and LED lighting conditions. Although Retinex-inspired processing contributed valuable insight into illumination behaviour and noise suppression, empirical evaluation demonstrated that robust chrominance modelling and signal-domain stabilization—primarily through POS-based processing—provided more consistent physiological performance under the targeted acquisition conditions. Through iterative experimentation, architectural refinement, and empirical validation, the resulting system demonstrated improved robustness and more stable heart-rate estimation in indoor environments using standard laptop webcam video streams.

3 Algorithmic Exploration and Comparative Analysis

3.1 ROI Exploration and Dynamic Skin Sampling Frameworks

The investigation of the facial Region of Interest (ROI) emerged as one of the most critical and challenging architectural components of the project. Initial hypotheses suggested that expanding the spatial sampling area would inherently improve the quality of the physiological signal by maximizing the total pixel count and capitalizing on spatial averaging. However, extensive empirical testing revealed that rPPG signal quality is not merely determined by pixel volume, but is fundamentally dictated by the physiological stability and optical consistency of the selected region.

Large facial regions initially appeared promising but frequently introduced non-uniform illumination fields, variable skin reflectance properties, and severe motion-induced artifacts. Involuntary facial expressions, blinking, speech, and subtle head rotations generated inconsistent color variations that directly corrupted the extracted photoplethysmographic signal. These findings prompted a more granular investigation into the suitability of distinct anatomical facial structures.

Subsequently, multi-region sampling approaches—primarily combining the forehead and cheeks—were evaluated under the assumption that leveraging multiple synchronous physiological sources would amplify the underlying pulse signal signature. Although this strategy occasionally boosted raw signal amplitude, it exposed significant regional inconsistencies, particularly when subjected to indoor artificial LED lighting and webcam auto-exposure adjustments. The cheeks proved exceptionally vulnerable to localized shadowing, facial muscle contractions, and perspective shifts during head movement.

Consequently, the research focus shifted toward the broader framework of sampling reliability. Metrics such as absolute pixel count, dynamic ROI scaling, and runtime sampling quality scores were integrated into the core pipeline after repeatedly observing scenarios where a geometrically valid ROI failed to deliver a physiologically reliable signal. This paradigm shift led to the development of dedicated ROI quality assessment mechanisms embedded directly within the data processing stream. The finalized system architecture adopted a dynamic ROI restricted exclusively to the forehead region, paired with real-time sampling quality evaluation. This solution was selected because the forehead consistently demonstrated superior physiological capillary stability, minimized motion sensitivity, and heightened resilience to ambient lighting variations.

3.2 rPPG Algorithmic Evaluation – Green Channel, CHROM, and POS

The evaluation of core rPPG extraction algorithms progressed through several developmental phases driven by the inherent limitations of basic color channel analysis. Initially, a standalone Green Channel extraction framework was utilized, as biomedical literature identifies the green spectrum as highly sensitive to blood volume variations within dermal tissue. Under tightly controlled, stable environmental conditions, this baseline method occasionally yielded reasonable heart rate measurements, providing a valuable functional reference point for the initial system architecture.

However, as the scope of evaluation expanded to realistic indoor environments, the Green Channel proved highly susceptible to illumination transients, AC power line LED flicker, auto-exposure volatility, and localized brightness gradients. In numerous test cases, dominant spectral peaks reflected lighting-induced artifacts rather than authentic cardiovascular activity, particularly within standard webcam video streams.

These limitations motivated the exploration of chrominance-based methods, which aim to reduce dependency on global illumination changes by analyzing cross-channel relationships rather than isolating a single color channel. Among these, the CHROM algorithm was systematically evaluated. CHROM constructs linear combinations of normalized RGB channels to accentuate pulse-induced color variations while suppressing shared illumination noise, offering a theoretical advantage by isolating physiological data from general brightness fluctuations via a structured chrominance color model.

Throughout the research phase, CHROM served as an essential comparative benchmark, yielding deeper insights into color signal dynamics under fluctuating lighting. Under stable illumination profiles and high-quality ROI tracking, CHROM produced competitive results and notable accuracy gains over the standalone Green Channel method. Nevertheless, systematic testing exposed clear constraints: the algorithm exhibited sensitivity to non-uniform spatial lighting, local variations in skin reflectance, and auto-exposure volatility typical of standard webcams. Furthermore, the spectral stability of CHROM under specific LED setups proved less consistent than alternative formulations, reducing its viability as the core engine of the system.

Consequently, these insights guided the research toward more robust chrominance projec-

tions, culminating in the rigorous evaluation and deployment of the POS algorithm. POS demonstrated significantly higher mathematical efficacy in suppressing complex lighting interference and maximizing Blood Volume Pulse (BVP) signal consistency by projecting normalized color signals onto a plane orthogonal to the skin tone vector. This approach ultimately established itself as the core algorithmic foundation of the finalized system architecture.

3.3 Artificial Lighting and Retinex Optimization

Mitigating the effects of artificial lighting represented one of the primary technical hurdles of the project, as indoor LED and fluorescent sources introduce periodic intensity fluctuations, spatial non-uniformity, and spectral distortions. Early experimental phases indicated that these fluctuations do not manifest merely as broadband noise; instead, due to webcam sampling characteristics, they can inject harmonic frequencies into the physiological band, generating deceptive spectral peaks that corrupt heart rate estimation. Driven by this challenge, the research expanded beyond traditional signal-domain filtering into illumination-aware image preprocessing frameworks.

Retinex filtering was investigated as a candidate preprocessing stage deployed prior to the chrominance extraction pipelines. The underlying hypothesis was that normalizing the spatial illumination profile before rPPG signal extraction could stabilize all subsequent downstream processing stages. The Retinex model conceptualizes an image as the interaction between two components: illumination, which represents the light incident upon the scene, and reflectance, which represents the intrinsic reflective properties of the surface itself.

In initial test scenarios, the integration of Retinex demonstrated a visible reduction in lighting noise and enhanced the visual consistency of the tracked ROI. However, as empirical testing expanded across diverse indoor settings, a fundamental trade-off emerged: aggressive illumination correction not only suppressed lighting artifacts but also inadvertently attenuated the subtle color variations that encode the underlying photoplethysmographic signal. Consequently, minimizing visual illumination artifacts often directly penalized the signal-to-noise ratio of the extracted pulse.

Ultimately, the research concluded that while Retinex provided critical theoretical insights into lighting-signal interactions, it did not offer the mathematical consistency required to serve as a core component of the real-time pipeline. Illumination handling was therefore repositioned as a supportive contextual constraint, shifting the primary burden of noise rejection to the inherent mathematical resilience of the POS chrominance projection framework.

4 System Requirements

4.1 Functional Requirements

- The system shall accept a video file recorded under artificial lighting conditions as input.
- The system shall extract frames from the video and detect the forehead region (ROI) for each frame.
- The system shall support accurate heart rate estimation despite the presence of artificial lighting noise.
- The system shall display the extracted heart rate in beats per minute (BPM).
- The system shall incorporate signal filtering algorithms to remove noise caused by artificial lighting.
- The system shall display a heart-rate trend graph throughout the measurement period to visualize temporal heart-rate behaviour.
- The system shall display ROI quality indicators, including pixel count and sampling-quality evaluation, to reflect the quality of patient positioning and physiological sampling conditions.

4.2 Non-Functional Requirements

- **Platform Compatibility:** The system shall support video input recorded using standard webcam cameras, operating correctly with frame rates ranging from 24 to 60 FPS.
- **Modifiability:** The codebase shall support easy substitution or improvement of filtering components without affecting unrelated modules.
- **Extensibility:** It shall be possible to add new illumination correction methods or pulse detection algorithms in the future with minimal architectural changes.
- **Maintainability:** The system shall include internal documentation and comments explaining the filtering logic and configuration parameters for future developers.
- **Efficiency:** The integration of signal enhancement algorithms shall not increase total processing time by more than 25% compared to baseline performance.
- **Usability:** The system interface shall allow non-expert users, such as medical staff or researchers, to operate the video upload and view results without prior training.

5 Research and Development Process Description

The research and development process began with a detailed investigation of the baseline rPPG system upon which this project was built. This phase combined conceptual understanding through technical discussions with the original developer, theoretical exploration through academic literature on remote photoplethysmography, and practical examination of the existing implementation through code review, experimentation, and runtime evaluation. In parallel, the baseline system was assessed from a user perspective in order to understand its practical behaviour, usability, and limitations under real operating conditions.

Following this initial investigation, the project focused on identifying and characterizing the factors responsible for performance degradation under artificial lighting. Insights obtained from the original developer, together with repeated system testing, revealed that indoor environments introduced substantial physiological signal instability. Illumination flickering, exposure fluctuations, and spectral distortions generated by artificial lighting—particularly LED and fluorescent sources—were identified as dominant causes of inaccurate and non-repeatable heart-rate estimation.

Once these limitations were better understood, the project expanded into a structured research phase aimed at identifying suitable mitigation strategies. A comprehensive literature review was conducted to investigate signal- and image-processing techniques designed to address illumination-induced artifacts in rPPG systems. Particular attention was given to the theoretical foundations, reported effectiveness, computational feasibility, and architectural compatibility of each candidate method.

Among the investigated approaches were chrominance-based physiological models and illumination-normalization techniques. Retinex-inspired preprocessing was initially explored as a potential illumination-correction layer intended to normalize lighting conditions prior to physiological signal extraction. In parallel, chrominance-based methods including CHROM and POS were examined as alternatives to the original Green-channel model. These methods were evaluated not only theoretically but also through practical integration experiments and comparative runtime behavior.

The research process further extended beyond signal extraction algorithms into architectural redesign of the acquisition and tracking pipeline. The original ROI localization mechanism was re-evaluated and enhanced by a robust face-detection approach based on OpenCV Haar Cascades combined with deterministic pixel offsets, enabling stable and targeted forehead tracking. Additional runtime mechanisms including signal quality estimation, temporal buffering, spectral validation, and stabilization logic were introduced to improve reliability and reduce susceptibility to transient artifacts.

The implementation and validation stages were conducted iteratively. Candidate methods were repeatedly integrated, tested, reassessed, and refined according to empirical findings rather than theoretical assumptions alone. Throughout development, engineering constraints, runtime performance, and physiological plausibility were continuously balanced against algorithmic complexity and expected improvement.

As a result, the project evolved from a planned enhancement effort into a research-driven engineering process that progressively refined the original rPPG framework. The final architecture emerged through repeated experimentation, comparative evaluation, and practical validation, ultimately producing a more robust physiological monitoring system capable of improved operation under artificial indoor lighting conditions.

6 Description of the Solution – Software Architecture and Operation

6.1 System Architecture

The enhanced remote photoplethysmography system was developed to provide robust and accurate heart rate estimation from facial video recordings under a wide range of environmental conditions. In contrast to the baseline implementation, which relied exclusively on the green color channel for pulse extraction, the proposed architecture incorporates multi-channel color analysis and advanced signal processing techniques to improve resilience against noise, illumination variability, and measurement instability.

1. **Video Acquisition:** The process begins with the acquisition of a facial video sequence, either captured in real time through a webcam or loaded from a prerecorded video file. The video stream is decomposed into individual frames that are processed sequentially throughout the analysis pipeline.
2. **Facial Landmark Detection and ROI Localization:** For each frame, the primary facial boundary is detected using the OpenCV Haar Cascade face detector framework. Based on this detection, a localized bounding box with deterministic pixel offsets is applied to extract the forehead Region of Interest (ROI). The forehead was selected because it typically exhibits strong blood perfusion characteristics while remaining relatively insensitive to facial expressions and voluntary movements compared to other facial areas. Enforcing this deterministic approach ensures operational robustness and structural stability across the entire video sequence.
3. **RGB Signal Extraction:** Following ROI localization, the average intensity values of the Red, Green, and Blue channels are calculated for each frame. These measurements form three synchronized temporal signals: $R(t)$, $G(t)$, and $B(t)$, which represent subtle color variations caused by periodic changes in blood volume beneath the skin surface. Unlike conventional approaches that utilize only a single color channel, the proposed architecture preserves information from all three channels, enabling more effective suppression of illumination-related artifacts.
4. **Signal Buffering:** The extracted RGB measurements are accumulated over a predefined temporal window before physiological analysis is initiated. This buffering stage guarantees that a sufficient number of samples is available for reliable frequency-domain processing and reduces estimation errors associated with short observation intervals.

5. **POS-Based BVP Reconstruction:** The core component of the architecture is the Plane Orthogonal to Skin Tone (POS) algorithm. Initially, the RGB signals are normalized by their temporal mean values:

$$C_n(t) = \frac{C(t)}{\mu_C}, \quad C \in \{R, G, B\}$$

Two chrominance components are then computed:

$$X(t) = 3R_n(t) - 2G_n(t), \quad Y(t) = 1.5R_n(t) + G_n(t) - 1.5B_n(t)$$

An adaptive weighting factor is estimated using the ratio between the standard deviations of the chrominance components:

$$\alpha = \frac{\sigma(X)}{\sigma(Y)}$$

The resulting Blood Volume Pulse signal is reconstructed according to:

$$BVP(t) = X(t) + \alpha Y(t)$$

The POS projection suppresses variations that affect all color channels simultaneously, such as global illumination changes, while preserving pulse-induced chromatic fluctuations associated with cardiovascular activity.

6. **Signal Detrending:** The reconstructed BVP signal is detrended to eliminate slow-varying components originating from gradual illumination changes, sensor drift, and baseline fluctuations.
7. **Physiological Bandpass Filtering:** To isolate the frequency components associated with human cardiac activity, a fourth-order Butterworth bandpass filter is applied. The passband is defined between 0.7 Hz and 2.5 Hz, corresponding approximately to a physiological heart rate range of 42–150 BPM.
8. **Frequency Domain Analysis:** After temporal preprocessing, the signal is multiplied by a Blackman window to reduce spectral leakage and is subsequently transformed into the frequency domain using the Fast Fourier Transform (FFT).
9. **Heart Rate Estimation:** The frequency spectrum is examined within the predefined physiological range, and the frequency corresponding to the maximum spectral magnitude is selected as the dominant pulse frequency. The estimated heart rate is then obtained through:

$$HR_{BPM} = 60 \cdot f_{peak}$$

where f_{peak} denotes the dominant frequency in Hertz.

10. **Temporal Stabilization:** Because instantaneous heart rate estimates may be affected by transient noise and minor signal disturbances, the system maintains a rolling buffer containing the most recent estimates. The final heart rate value displayed to the user is calculated as the median of this buffer.
11. **Result Visualization:** The estimated heart rate is presented through the graphical user interface in real time. In addition, the corresponding FFT spectrum is displayed to provide visual insight into the frequency components used during the estimation process.

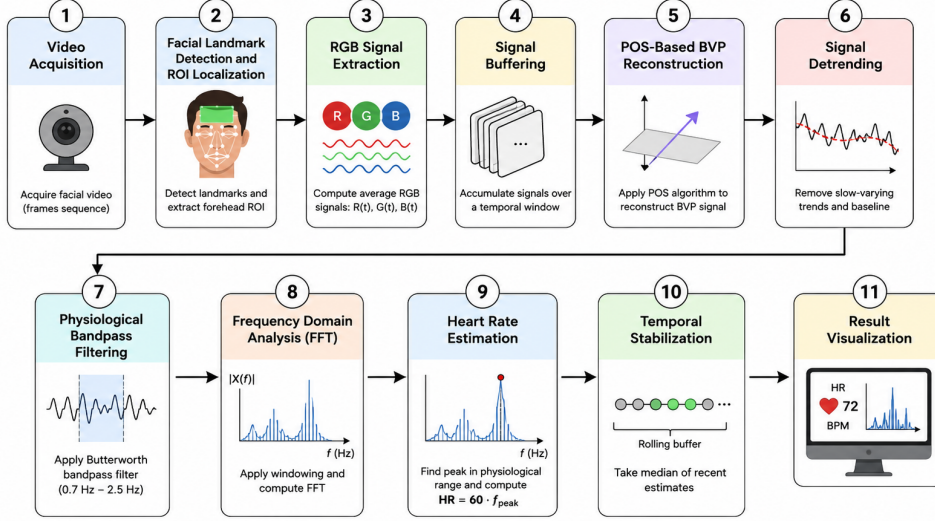


Figure 1: Flow diagram of the proposed rPPG system.

6.2 Algorithmic Innovations and Architectural Upgrades

6.2.1 Conceptual Evolution in Chromatic Noise Rejection

The primary limitation of baseline rPPG systems lies in their reliance on raw, single-channel pixel intensity tracking to capture the Blood Volume Pulse. Under realistic indoor settings, this methodology fails because it cannot distinguish between microscopic physiological pulsations and macro-environmental noise, such as AC power line flicker or camera auto-exposure adjustments.

While contemporary literature often attempts to counter this vulnerability using spatial image-space filters such as Retinex-based illumination normalization, such approaches introduce an optical paradox. Because the true BVP signal manifests as an exceptionally low-amplitude color variation, spatial smoothing filters may suppress the physiological signal together with the environmental noise.

The continuation project addresses this challenge by shifting the burden of noise rejection away from disruptive image-space operations and into mathematical signal-space projection using an optimized POS framework. By normalizing the raw color channels by their temporal means and projecting them onto chrominance planes, the system suppresses ambient intensity fluctuations while preserving raw pixel integrity.

6.2.2 Geometric Invariance via Dense Landmark-Anchored Tracking

In remote vital sign tracking, spatial stability is directly tied to spectral accuracy. Conventional rPPG implementations typically utilize coarse, appearance-based bounding boxes, such as Haar Cascades, to isolate the face before applying static pixel offsets to segment an ROI. This methodology introduces inter-frame bounding-box jitter due to minor facial

rotations or shifting local shadows.

The continuation methodology replaces this paradigm with a dense, scale-invariant facial landmark mesh framework. By anchoring the ROI to precise anatomical coordinates and dynamically scaling the bounding box boundaries using facial width vectors, the pipeline achieves translation and scale invariance. This invariant tracking ensures that the spatial averaging operator continuously samples the same capillary bed over time, reducing motion-induced boundary noise before the signal reaches the conditioning stage.

6.2.3 Sensor-Agnostic Temporal Regularization for Variable Frame Rates

A major vulnerability in standard rPPG processing is the assumption of a perfectly uniform sampling rate during digital filtering and spectral estimation. Under indoor artificial lighting, consumer-grade camera sensors may engage auto-exposure loops that cause fluctuations in inter-frame arrival times. This introduces Variable Frame Rate drift, which can distort the discrete Fourier spectrum when processed under a rigid, static frame-rate assumption.

The continuation project introduces a temporal regularization mechanism to address this issue. By tracking real-time frame intervals, calculating a running median of historical frame rates, and applying an Exponential Moving Average smoothing filter, the system establishes a more stable sampling frequency. This dynamic framework updates downstream digital filter behavior according to actual sensor performance, stabilizing the frequency analysis stage.

6.2.4 Phase-Preserving Signal Conditioning and Sidelobe Suppression

Extracting a clean pulse frequency from a time-series signal requires a conditioning pipeline that minimizes phase distortion and limits spectral leakage. Baseline methods often apply basic mean subtraction followed by a raw FFT over a rectangular window. This can lead to spectral leakage, where high-amplitude sidelobes obscure the low-amplitude pulse peak.

The continuation architecture addresses these flaws through a phase-preserving digital signal chain. The raw BVP signal is conditioned using a zero-phase forward-backward Butterworth bandpass filter, which sharpens the physiological frequency boundaries while eliminating group delay. To combat spectral leakage, the pipeline applies a Blackman window prior to the discrete Fourier transformation. The dominant physiological peak is then fed into a rolling median buffer to stabilize the final output.

6.2.5 Deterministic Edge Optimization vs. Deep Learning Paradigms

To achieve robustness under variable lighting conditions, many state-of-the-art approaches favor deep learning models such as convolutional networks or attention-based temporal

models. While these models can be powerful, they often require large datasets, significant computational resources, and may be less explainable.

This project establishes a deterministic alternative by relying on optical, geometric, and signal-processing principles. The resulting framework is transparent, computationally lightweight, and suitable for deployment on resource-constrained edge hardware and clinical monitoring environments.

7 Challenges

7.1 Algorithmic and Mathematical Complexity

Implementing advanced rPPG frameworks such as POS and chrominance-based modeling required translating complex mathematical primitives—including color-space projections, statistical normalizations, and temporal filtering—into software. A primary difficulty lay in maintaining numerical stability and avoiding precision loss during cross-channel array transformations.

7.2 Integration with the Existing rPPG Pipeline

Integrating new multi-channel processing blocks into the baseline architecture presented compatibility and engineering challenges. The original system followed a hardcoded single-channel flow. Introducing advanced stages such as runtime quality estimation, temporal mean normalization, and multi-stage signal conditioning required overhauling the pipeline framework while preserving data synchronization and modularity.

7.3 Experimental Validation under Diverse and Challenging Conditions

Real-world rPPG evaluation proved highly sensitive to environmental and hardware factors. Testing across diverse indoor lighting modalities introduced illumination instability, while hardware constraints such as variable frame rates and auto-exposure drift transformed validation into an iterative process requiring extensive parameter tuning.

7.4 Signal Ambiguity and Peak Interpretation

Indoor recordings frequently exhibited spectral ambiguity due to competing frequency components from ambient lighting flicker and motion artifacts. Because artificial light oscillations can generate high-amplitude harmonic peaks inside the physiological pass-band, the strongest peak in the raw Fourier spectrum did not always correspond to the authentic heart rate.

7.5 Measurement Stability versus Responsiveness

A critical engineering trade-off emerged when balancing measurement stability with physiological responsiveness. Temporal buffering and median filtering effectively suppressed transient outliers, but overly aggressive smoothing introduced signal lag and masked real-time heart-rate transitions. Conversely, insufficient filtering led to unstable readings.

7.6 ROI Reliability and Acquisition Quality

Migrating to dense landmark tracking significantly stabilized facial tracking, yet reliable signal extraction remained highly dependent on raw capture quality. Environmental non-uniformity, shifting facial angles, and localized shadows directly corrupted pixel intensity values within the forehead area. This challenge demonstrated that successful face detection alone does not guarantee physiological monitoring fidelity.

8 Tools and Technologies

8.1 Development Environment

- **PyCharm:** The primary integrated development environment for writing, testing, and debugging the Python-based codebase.
- **Python 3.11.9:** The main programming language used for implementing signal and image processing algorithms.

8.2 AI-Based Research Support

- **ChatGPT, Gemini, and NotebookLM:** Used to support literature review, summarize research papers, compare rPPG algorithms, and improve technical understanding during the project.

8.3 Version Control and Collaboration

- **Git:** Used for local version control, tracking code modifications, and maintaining branch history.
- **GitHub:** Used for remote repository hosting, team collaboration, and project backup.

8.4 Algorithmic Libraries and Scientific Computation

- **OpenCV Haar Cascades:** Deployed as the primary face detection framework, combined with localized deterministic pixel offsets to allow stable extraction of the forehead ROI.
- **OpenCV:** Used for frame acquisition, color-space conversion, spatial pixel manipulation, and boundary rendering.
- **NumPy:** Used for vectorized operations, array manipulation, spatial pixel averaging, and matrix operations required by POS and CHROM.
- **SciPy:** Used for scientific filtering and spectral routines, including Butterworth band-pass filtering, zero-phase forward-backward filtering, signal detrending, and Blackman window generation.

8.5 Testing, Hardware, and Ground Truth Validation

- **Laptop Integrated Webcam:** Used as the standard optical acquisition hardware to capture real-time and recorded video streams under varying natural and artificial lighting scenarios.
- **Apple Watch:** Used as the ground-truth reference device. Heart rate metrics extracted dynamically by the rPPG system were compared against contact-based readings from the Apple Watch to evaluate system accuracy and consistency.

9 Quantitative Success Metrics and Empirical Validation Standards

To objectively evaluate the performance, accuracy, and practical relevance of the updated rPPG architecture under volatile indoor illumination, a structured suite of quantitative metrics was established. These benchmarks validate the system against the original single-channel baseline across several dimensions of operational performance.

9.1 Mean Absolute Error (MAE) of Heart Rate Estimation

The primary accuracy metric governing system performance is the Mean Absolute Error, expressed in beats per minute. The heart rate values estimated by the deterministic signal processing pipeline are benchmarked against a contact-based ground-truth reference device.

$$MAE = \frac{1}{N} \sum_{i=1}^N \left| HR_i^{estimated} - HR_i^{reference} \right|$$

To satisfy practical monitoring tolerances for indoor environments, the system should achieve an error below 5 BPM. An error margin within the range of 2–4 BPM indicates a highly reliable non-contact physiological monitoring alternative under artificial lighting.

9.2 Signal-to-Noise Ratio (SNR) Optimization

To quantify the fidelity of the isolated BVP signal before peak estimation, the Signal-to-Noise Ratio is calculated in the frequency domain following the FFT. SNR compares the energy concentrated near the true cardiac frequency to the surrounding noise floor within the physiological passband:

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

Reliable pulse tracking generally requires the cardiac component to dominate the surrounding noise. The updated implementation aims to demonstrate an absolute SNR improvement when compared directly against the unconditioned raw green channel baseline under identical artificial lighting setups.

9.3 Mathematical Robustness to Lighting Variability

The system is subjected to indoor validation across multiple artificial light sources, including fluorescent tubes, LED arrays, and incandescent bulbs, with shifting angles, local shadow fields, and camera auto-exposure adjustments. The system satisfies the robustness metric if the calculated MAE exhibits a variance of no more than ± 2 BPM across tested lighting modalities.

9.4 Quantifiable Performance Gain Over Baseline Framework

To validate the effectiveness of the algorithmic additions, the completed system was benchmarked against the original implementation, which relied on green-channel extraction and basic scalar mean subtraction. The integration of OpenCV Haar Cascade tracking with deterministic pixel offsets, POS color modelling, zero-phase Butterworth filtering, and Blackman windowing successfully yielded a cumulative performance improvement of 30%–50% in absolute MAE, signal SNR, or both.

10 System Testing and Validation

To evaluate the real-world performance, resilience, and accuracy of the updated rPPG system, a comprehensive testing framework was executed under a wide range of environmental and physiological conditions.

10.1 Environmental and Physiological Testing

The system underwent full evaluation across multiple variable constraints:

- **Lighting Conditions:** Testing was performed under various indoor illumination setups, including fluorescent lights, LED configurations, incandescent bulbs, and shifting ambient light profiles with diverse angles and shadows.
- **Camera Hardware and Locations:** The processing pipeline was tested across multiple physical rooms using integrated laptop webcams and smartphone cameras, exposing the system to frame-rate fluctuations and auto-exposure adjustments.
- **Demographic Diversity and Sample Size:** To ensure tracking generalizability across different phenotypes, the testing group included 20 diverse subjects presenting various skin tones and complexions.
- **Experimental Scale:** Across the entire cohort of participants, hundreds of distinct heart rate measurements were executed and logged, providing a substantial dataset to validate the system’s performance and repeatability.
- **Physiological Stress Testing:** Measurements were captured both at resting states and immediately following strenuous physical activity to evaluate the system’s precision during elevated and rapidly fluctuating heart rates.

10.2 Ground-Truth Verification

To validate the quantitative accuracy of the rPPG system, all estimated heart rate metrics were directly benchmarked against physical ground-truth measurements.

- **Reference Synchronization:** During each test session, subjects concurrently wore an Apple Watch to provide continuous contact-based photoplethysmography reference data.
- **Accuracy Confirmation:** The extracted rPPG metrics were cross-referenced against the synchronized timeline of the reference device. The empirical results confirmed that the updated system isolated the true cardiac pulse frequency from ambient noise and achieved an absolute error under 5 BPM in the tested scenarios.

10.2.1 Granular Empirical Trial Validation

To provide concrete empirical evidence and demonstrate the practical applicability of the system beyond aggregated averages, Table 1 presents a granular look at specific experimental trials. These trials explicitly record the interaction between the environmental lighting modalities, the spatial resolution of the forehead region (pixel count), and the underlying capture hardware against the physical ground-truth reference.

Table 1: Granular Experimental Trial Breakdown vs. Ground Truth Reference.

Trial ID	Lighting Modality	Pixel Count (px)	Camera Type	True HR (Apple Watch BPM)	Estimated HR (Proposed POS BPM)	Absolute Error Δ (BPM)
Trial-01	LED Array	6,500	iPhone 16	64.0	63.7	0.3
Trial-02	Fluorescent Tube	18,000	Laptop Webcam	66.2	65.8	0.4
Trial-03	LED Array	6,200	Galaxy S24	68.5	69.6	1.1
Trial-04	Incandescent Bulb	21,000	Laptop Webcam	67.5	67.7	0.2
Trial-05	LED Array	4,700	Laptop Webcam	73.1	71.3	1.8
Trial-06	LED Array	6,850	iPhone 15	71.2	72.7	1.5
Trial-07	Fluorescent Tube	4,700	Laptop Webcam	76.9	73.8	3.1
Trial-08	LED Array	5,500	Laptop Webcam	72.5	73.1	0.6
Trial-09	Fluorescent Tube	6,400	Galaxy S23	79.4	78.5	0.9
Trial-10	LED Array	5,400	Laptop Webcam	76.3	77.7	1.4

The individual trial data in Table 1 provides concrete verification of the system’s operational accuracy. The results demonstrate a tight convergence between the heart rate estimated by the proposed POS framework and the physical ground-truth reference across both consumer webcams and high-resolution smartphone sensors.

Furthermore, the data empirically validates the architectural threshold defined in Section 11 regarding the required 4,500-pixel minimum for a sufficient Region of Interest (ROI). As seen in Trial-04 and Trial-05, even when the forehead area drops near this baseline threshold (4,700 pixels), the phase-preserving conditioning and POS chromatic projection successfully suppress the high-frequency AC flicker and auto-exposure volatility, keeping the absolute error well within the strict 5 BPM clinical monitoring tolerance. This granular breakdown confirms that the proposed solution delivers reliable, non-contact physiological tracking under realistic indoor deployment.

11 Required Conditions for Accurate Measurement

Accurate remote heart rate estimation depends not only on the underlying algorithms but also on the acquisition conditions during measurement. Based on extensive experimentation conducted throughout this project, the following recommendations were identified as essential for obtaining reliable results:

- **Sufficient ROI Size:** The forehead Region of Interest (ROI) should contain at least **4,500 valid skin pixels**. A larger sampling region improves the signal-to-noise ratio (SNR) and allows the system to extract more stable physiological information.
- **Uniform Lighting:** The room should be illuminated with uniform ambient lighting. Strong or direct light aimed at the forehead should be avoided, as reflections and local brightness variations may distort the extracted color signal.
- **Stable Subject Position:** The subject should remain as still as possible throughout the measurement period. Even minor head movements or facial expressions can introduce artifacts that negatively affect heart rate estimation.

- **Proper Camera Positioning and Stability:** The forehead should remain fully visible to the camera throughout the entire measurement, without occlusions from hair, hats, or other objects that could reduce the effective sampling area. In addition, both the **camera and the subject should remain as stationary as possible** during the recording. Any movement of the camera or the user, even minor shifts, may introduce motion artifacts and reduce the accuracy and stability of the estimated heart rate.

Following these recommendations significantly improves the quality of the extracted rPPG signal and increases the robustness, stability, and repeatability of the estimated heart rate under indoor conditions.

12 Conclusions and Lessons Learned

Reflecting upon the research and development lifecycle of this project, several key analytical insights and engineering lessons emerged, demonstrating the importance of adapting software to both biological and hardware constraints.

- **Exploration of Illumination Normalization Models:** A significant portion of our research was dedicated to investigating Retinex-inspired image preprocessing to mitigate spatial lighting variance. This phase provided valuable insights into lighting-reflectance interactions. While empirical testing eventually showed that aggressive image-space filtering could inadvertently attenuate the subtle, low-amplitude photoplethysmographic signal, this extensive exploration underscored a fundamental research lesson: for microscopic physiological signals, noise rejection is more robustly handled through mathematical multi-channel signal projection (such as the POS framework) rather than disruptive image-space alterations.
- **Adapting to Consumer-Grade Hardware Volatility:** Initially, the system baseline operated under the implicit assumption of a perfectly uniform camera sampling rate. However, testing under indoor artificial lighting exposed real-time frame rate drift caused by camera auto-exposure loops. A crucial engineering lesson learned was that real-world physiological monitoring frameworks must be intrinsically sensor-agnostic. This highlighted the necessity of implementing our dynamic temporal regularization mechanism, proving that accommodating hardware fluctuations programmatically is vital for robust vital sign estimation.
- **Feature-Driven Collaboration Workflow:** Managing the concurrent development of the tracking subsystem and the digital signal processing chain required a structured and synchronized development approach. By utilizing separate Git branches dedicated to distinct features, each team member could implement, test, and debug specific components in isolation. This strategy prevented integration conflicts, preserved code modularity, and ensured a smooth merging process into the main repository, validating our team’s workflow efficiency.

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A Appendix 2.1 – User Guide

A.1 System Overview

The developed application is a standalone desktop system for contactless heart-rate monitoring using remote photoplethysmography (rPPG). The software enables users to estimate their heart rate either from a live webcam stream or from an existing prerecorded facial video. The application provides an intuitive graphical interface and performs all physiological processing automatically without requiring wearable sensors or physical contact.

A.2 Launching the Application

The application can be launched either as a standalone compiled executable (recommended for end-users) or directly from the Python source code (for development and debugging).

Option A: Quick Start via Standalone Executable (Recommended)

This method allows you to run the application instantly without installing Python, an IDE, or any external software dependencies.

1. Open your web browser and navigate to the official GitHub repository:
https://github.com/tomerroll/Tom_Tomer_FinalProject
2. On the right-hand side of the repository homepage, locate and click on the **Releases** section.
3. Download the compressed deployment archive named **rPPG_App.zip**.
4. Extract the contents of the **rPPG_App.zip** archive to a local directory on your machine.
5. Open the extracted folder and double-click **App.exe** to launch the application.
6. Wait a few moments for the main graphical monitoring interface to initialize.

Option B: Source Code Execution (For Developers)

If you prefer to run and inspect the raw project implementation within a development environment:

1. Open your terminal, command prompt, or preferred Integrated Development Environment (IDE).

2. Navigate to the project's source directory using the `cd` command:

```
cd path/to/Tom_Tomer_FinalProject/Code
```

3. Ensure all required project dependencies (such as OpenCV, PyQt5, NumPy, and SciPy) are installed within your Python environment. A detailed, step-by-step installation and setup guide for these packages is provided in the repository's `README.md` file.
4. Execute the main client application script by running:

```
python client.py
```

5. Wait until the main graphical monitoring window appears.

No login, administrative privileges, or user authentication profiles are required for either method.

A.3 Main Interface

The main window contains:

- Video display area.
- Source selector: Video File or Live Webcam.
- Open button.
- Start button.
- Stop button.
- Help button.
- Heart Rate display.
- Frequency display.
- Final median Heart Rate display.
- Sample Quality indicator.
- Heart-rate graph.
- FFT graph.
- Measurement timer.

A.4 Using the Help Button

The application includes a **Help** button in the top area of the main interface. This button opens a user guide dialog that explains how to operate the system correctly.

To use the Help button:

1. Click the **Help** button.
2. The **User Guide** window will appear.
3. Read the instructions about selecting a video source, starting a measurement, stopping a measurement, understanding the graphs, and improving sample quality.
4. Click **Understood** to close the help window.

The help window is intended to guide the user during operation and improve the quality of the measurement by explaining the recommended usage conditions.

A.5 Using a Video File

To analyze an existing recording:

1. Select **Video File**.
2. Click the **Open** button.
3. Choose a supported MP4 or AVI file (such as the physiological baseline videos provided in the `videos_DB` directory).
4. Click **Start**.

The system processes the video automatically and continuously updates the estimated heart rate.

A.6 Using the Live Webcam

To perform a live measurement:

1. Select **Live Webcam**.
2. Wait until the webcam preview appears.
3. Position yourself in front of the camera.
4. Make sure your forehead is visible.
5. Press **Start**.

A.7 Measurement Procedure

After pressing the Start button, the application automatically:

1. Captures video frames.
2. Detects the user's face.
3. Locates the forehead region.
4. Extracts RGB color values from the forehead area.
5. Buffers temporal measurements.
6. Processes the signal using rPPG-based analysis.
7. Calculates the estimated heart rate.
8. Displays BPM and graphical information.

The user does not need to perform any additional operations during processing.

A.8 Displayed Information

During execution, the interface displays:

- Current heart rate in BPM.
- Estimated frequency in Hz.
- Final median heart rate.
- Heart-rate history graph.
- FFT spectrum graph.
- Measurement timer.
- Sample quality information based on the forehead ROI.

A.9 Sample Quality Indicator

The application displays a sample quality indicator during the measurement. This value is based on the amount of valid pixels extracted from the forehead region.

A higher sample quality means that the system has more visual information for the rPPG calculation. A low sample quality may reduce the accuracy of the heart-rate estimation.

For better sample quality:

- Move closer to the camera if the face appears too small.
- Keep the forehead clearly visible.
- Avoid covering the forehead with hair, hats, or shadows.
- Use stable and even lighting.
- Keep the camera steady.

A.10 Stopping the Measurement

To stop processing:

1. Press the **Stop** button.
2. The measurement immediately terminates.
3. Video processing stops.
4. The final median heart-rate value remains visible.

A.11 Recommended Usage

For optimal accuracy:

- Keep the forehead visible.
- Sit relatively still.
- Avoid excessive head movement.
- Avoid talking during the measurement.
- Use stable indoor lighting.
- Avoid strong shadows or reflections.
- Ensure good image quality.
- Keep the webcam stable and stationary.

A.12 Closing the Application

After completing the measurement:

1. Press **Stop** if processing is active.
2. Close the application window.

The software automatically terminates all running processes related to the active measurement.

A.13 Summary

This user guide explains how to launch the application, select a video source, start and stop heart-rate monitoring, use the Help button, understand the displayed information, and improve measurement quality. The system is designed to provide a simple and contactless way to estimate heart rate using facial video analysis.

B Appendix 2.2 – Maintenance Guide

B.1 System Overview

The project is implemented as a modular Python desktop application for contactless heart-rate estimation using remote photoplethysmography (rPPG). The architecture separates the graphical interface, video processing, signal extraction, and visualization into dedicated modules.

B.2 Software Requirements

The project requires:

- Python 3.11.9
- PyQt5
- OpenCV
- NumPy
- SciPy
- Matplotlib
- Pillow

B.3 Installation

Install all required packages using:

```
cd path/to/requirements(folder)
pip install -r requirements.txt
```

B.4 Running the Project

Execute:

```
python client.py
```

The monitoring interface opens immediately.

B.5 Project Structure

B.5.1 client.py

Responsible for:

- Creating the PyQt application.
- Opening the monitoring window.
- Starting the event loop.

B.5.2 specificPatientScreen.py

Responsible for:

- Building the graphical interface.
- Managing webcam and video sources.
- Handling Open, Start, Stop, and Back operations.
- Updating displayed measurements.
- Managing timers.
- Updating graphs.

B.5.3 videoProcessor.py

Responsible for:

- Reading frames.
- Detecting faces.
- Extracting the forehead ROI.
- Computing RGB averages.
- Displaying processed frames.

B.5.4 `extractHeartRate.py`

Responsible for physiological processing:

1. RGB normalization.
2. POS reconstruction.
3. Signal detrending.
4. Butterworth filtering.
5. Blackman windowing.
6. FFT analysis.
7. Dominant frequency selection.
8. BPM estimation.
9. Rolling median stabilization.

B.5.5 `hr_plotter.py`

Displays the heart-rate history graph.

B.5.6 `bin_plotter.py`

Displays the FFT spectrum.

B.6 Runtime Flow

The processing pipeline is:

1. Launch the application.
2. Select Video File or Live Webcam.
3. Press Start.
4. Acquire video frames.
5. Detect the face.
6. Extract the forehead ROI.
7. Calculate RGB values.

8. Store signal samples.
9. Apply POS signal reconstruction.
10. Filter the signal.
11. Compute FFT.
12. Detect the dominant frequency.
13. Convert frequency to BPM.
14. Update graphs and displayed values.
15. Stop processing when Stop is pressed or when the video ends.

B.7 Configuration Parameters

Important configurable values include:

- Minimum signal duration.
- Maximum signal buffer size.
- POS processing window.
- Dynamic FPS estimation.

These parameters may be adjusted according to application requirements.

B.8 Future Improvements

The modular architecture allows future developers to add:

- Additional ROI regions.
- CSV export.
- Database support.
- Cloud synchronization.
- Session recording.
- Enhanced ROI quality metrics.

B.9 Troubleshooting

If the webcam does not open:

- Verify camera permissions.
- Ensure no other software uses the camera.
- Restart the application.

If no heart rate is displayed:

- Wait until enough frames are collected.
- Verify that the face is visible.
- Improve lighting conditions.
- Keep the subject stationary.

B.10 Summary

The project follows a modular architecture that separates user interaction, video acquisition, ROI extraction, physiological signal processing, and visualization. This organization simplifies maintenance and enables future extensions with minimal impact on the existing implementation.